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Point Process based time sensitive personalised recommendation

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Abstract

We frequently rely on suggestions in an online context, including those from search engine results, e-commerce product recommendations, movie recommendations, and so forth. These suggestions are made in response to our actions, such as when we search for products, songs, or movies, or at the very least, when we visit the website in order to receive a recommendation. These recommendations do not take into account time-sensitive factors, such as when we often utilise these services during the day or the week. We prefer to repeat some activities, like listening to music at a specific time of day. We presented a methodology in this paper that addresses two critical issues in personalised recommendation. 1) recommending the right items to the right people at the right time, and 2) estimating when a user will return to a service or product after performing repeated actions. The scholarly community has not yet examined this work in personalised recommendations. To do this, we offered the Hawkes process, in which we employed a customised initial intensity based on Hierarchical Poisson Factorization, and for a dynamic activity, we considered a sinusoidal function coupled with exponential effect decay. This is consistent with user activity cycles, such as music listening during a specific time of day. This is the first framework we've used, which is based on a probabilistic matrix factorization.

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1. INTRODUCTION

Social networking and e-commerce are developing at such a fast rate that they will eventually permeate everyone's daily lives. People find it comfortable to rely on it to save their time and energy in a busy lifestyle. For instance, Siri, Facebook, OLX, Alibaba, and Amazon. According to the Fleder Blockbuster[1], recommendation systems are crucial to user engagement. Online social networks and management platforms like QQ Music, Baidu Music, and Youtube constantly garner a sizable user base. A fundamental and significant goal of these services is to provide suggestions for appropriate items. These online platforms can improve user fulfilment and boost user engagement or website income

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by recognising users' wants and providing them with potentially exciting items because of expanded user engagement, item buys, virtual transactions, as well as ad-clicks [2, 3].

Recommending a personalised user experience is playing an essential role in engaging users in online services [2]. For instance, making personalised suggestions on the correct item at the correct time can make the personalised recommendation more beneficial for producer and the consumer. In our daily life schedule, we all tend to sync with time on some activities, which could be daily; some may be weekly or even monthly. Which is also true for consuming online and offline products, such as listening to the same music, purchasing the same grocery items on similar days of the month or weeks, and so on. Sadly, the most existing personalised recommendations are based on one-time recommendations only; they don't consider users' repetitive behavior. Existing systems are streamlined at foreseeing users' one-time inclinations (frequently indicated by whole number evaluations) on things, while users' constantly time-varying inclinations remain to a great extent under-explored. It is found [4, 5] users' interests and items' features evolve over time, which can trigger unexpected interaction. The interaction between user and item over time evolves, which results in dynamic user interests and also item features. For example, a new disease can be triggered by a previous long term disease or people can develop new interests over time [6]. The popularity of items among different targeted groups leads to redefining the item feature movie, such as how kids' movies for kids can gain popularity among adults.

It is found that the user tends to consume items according to recommendation [1]. But in the previous research on recommendation, it has not been explored in detail for recommending the right item to the right person and at the right time [7]. Especially items which user repeatedly consumed over time [8, 6]. It allows successful prediction of the return time of the user to the services. Some researchers presented a framework for predicting returning time where recurrent neural networks were exploited along with a survival loss function [9]. It is also found that conventional user feedback such as user given ratings, click-through rates, and so on is found to be ineffective for measuring user engagement due to sparse and noisy dataset [2]. Therefore, researchers found temporal features to be effective, such as when a user comes again to the service [10].

Moreover, effective predictions of the returning time of a user to the service will help to improve marketing techniques such as providing ad-words at the right time by allowing marketers to bid using time slots. Another time-sensitive prediction helps in knowing the evolving user and item choices. With regards to current electronic health services, patients may have a few diseases that have a complex dependency on one another. The event of one disease could initiate another disease. Anticipating the returning time on specific illness can adequately assist doctors with taking proactive actions to decrease the potential dangers of triggering another disease [11, 6]. However most of the previous works have been focused on predicting explicit ratings [12, 13, 14] but very few on predicting returning behavior of users. Therefore exploring this direction of research has practical significance.

1.1. Research problem

Existing research solely takes into account users' preferences as being static, ignoring the fact that different users' preferences change over time. The primary hypothesis of this research is that users' repetitious behaviour is not taken into account by the previous research. This study suggests creating a personalised recommendation algorithm that takes into account users' and products' changing preferences as well as their frequent consumption patterns. In order to take into account consumers' changing choice predictions, this study suggests point process-based models. As outlined below, we shall address the following challenges:

1.1.1. Predicting right item to right user

Taking into account both the features of the thing and how online users choose them. Online users' preferences can change over time; for instance, a comedy lover may discover some fantastic action movies and start to like them. Unfortunately, current technologies do not take user behaviour changes into account. The characteristics of goods can also alter over time in a similar way. These two aspects also need to be taken into account in the current investigation. Studies have shown that when products are recommended, users' preferences may alter since the quality of the products depends on how often they are used. Any model that is based on collaborative filtering will be used to anticipate the appropriate item for the appropriate user. More specifically, we will use any low rank based model as a basis, such as HPF, DPF, [5] and so on. Furthermore, whenever we recommend an item list to user u at time t , we will sort the recommended item according to the intensity described in the proposed scheme section.

1.1.2. Modeling recurrent consumption behaviour

The fact that some products, like songs and groceries, are consumed repeatedly by users is trivial. However, there is a pattern to this consumption, such as a particular time of day, week, or even month. We will model the time-varying and recurrent customers' taste problems based on actual findings. Following a thorough examination of the item consumption patterns of each user, this research proposed a model that will yield superior results. The sinusoidal recurrent kernel we have presented for the current model will successfully describe the periodic and recurring behaviour of individual users. It is clear from Fig. 1 how our suggested model will suggest giving recurrent items priority. In the figure, users show recurrent item consumption behaviours such as listening, reading, and eating, though at the same time it will also consider the collaborative filtering effect. So in Fig. 1 the first recommendation is because of recurrent behaviour of user while second is because of collaborative filtering with user u_2 .

1.1.3. Predicting returning time of user to a service

The traditional static recommendations are based on similarity in how users and items are related and therefore give a one-time recommendation. The presence of a huge amount of data which users leave while interacting with the item in an online environment gives opportunity to researchers. From this huge and precise data, the individual users' behaviour can be learned efficiently and thereby enhance the user experience. The user transaction data from user activity and the returning time of the user to a service should be modeled. In the presence of the latest technologies such as deep learning, some researchers have tried to mine those implicit behaviours and recommend time sensitive recommendations. Researchers in [15] used recurrent neural networks to give session based recommendations. Along with recurrent neural networks, some researchers [16] exploited point process methods for predicting recurrent events. Some researchers [10] predicted the returning time using a hazard based model, namely the Cox proportional Hazard model. As previous traditional models ignore the returning time prediction, this research will study the returning behaviour of the users from the data and correspondingly model it. In this research, we will use a point process based model to predict the next returning time of a user. From our framework, we will predict the returning time of a user by predicting the next recurrent time by some state-of-the-art method such as that described in the proposed scheme section.

The structure of this paper is as follows. The background is discussed in section [2] after the introduction. In section [1.1] we have outlined the research content. More specifically, we have discussed how data was gathered and used in this study in section [4.1]. Section [4.2] describes the evaluation metric that was employed to gauge the effectiveness of the suggested algorithm. Section [3] contains a description of the proposed model. Section [] displays the experimental results. Finally, the work has been finished in section [5].

2. LITERATURE REVIEW

Traditionally, recommendation problems have revolved around predicting explicit rating problems [17, 18]. Although temporal effects were ignored for a long time. Even if some researchers considered the temporal effect, they did not consider the future evolution of the user item interaction, i.e. they were only able to predict a fixed rating for an item in the future. [19, 4, 20]. The static nature of prediction, which is what has been researched in the last decades, ignores the varying user tastes over time. Until recently, researchers gave attention [2, 6, 8, 21, 22, 23]. Considering users' varying tastes was also hard since we didn't have the realisation of future varying interests as well as a lack of explicit data until some researchers gave some sophisticated algorithms. The lack of explicit rating problems was solved by considering implicit user feedback, which can be extracted from users' transaction logs. The models that consider implicit user feedback are considered better than those that are based on explicit user ratings. This gives us another clue for dynamically modelling user interests because those data are abundant in comparison to explicit user data (rating).

Low-rank matrix factorization has been traced long back since its inception for Principal Component Analysis (PCA) [24]. In which the target was to extract minimum dimensions from higher dimensions, such as data representation loss, which is the minimum. In the late 20th century, we realised its importance when we had its application in data analysis when there was an abundance of dimensions. This area is getting attention in different forms, such as non-negative matrix factorization (NMF). It was first developed for factorising matrices of image recognition [25]. Although the concept of non-negativity originated in cerebral representation, it now has a wide range of applications,

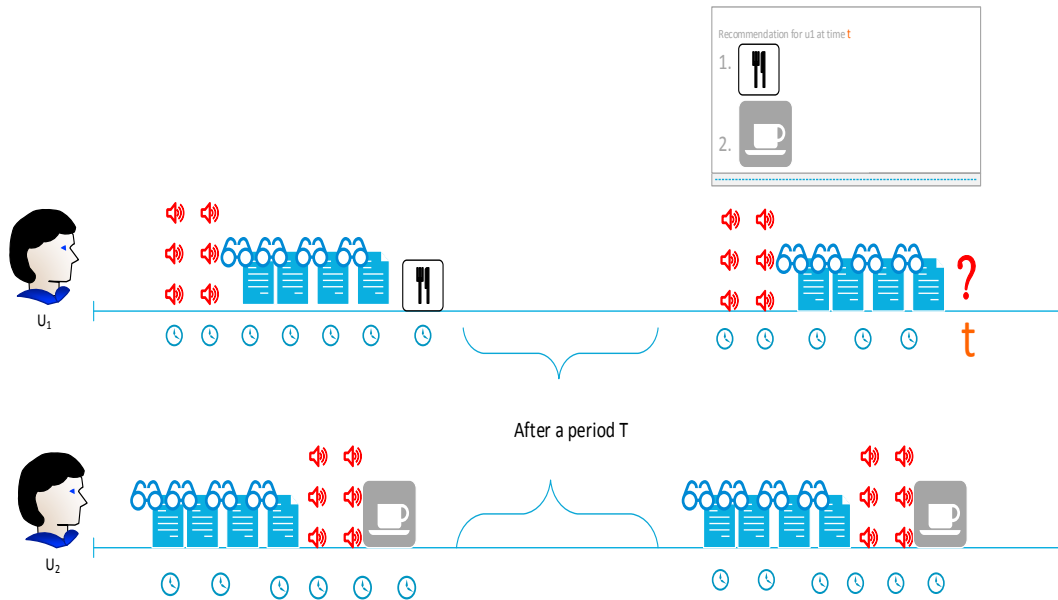


Fig. 1. This figures shows how our proposed model will consider both personalised as well as time varying effect in recommendation to user u_1 .

including recommendation systems, image recognition, dimensionality reduction, clustering, and so on. One of the well-utilized areas is user-item sparse matrix factorization in recommendation systems. As in real time data, there are too few ratings by users and very few rated items. In collaborative filtering, two theories are applied; the first one considers the value is missing at random [26] and the second one considers the non-missing entries follow some distribution [27].

Poisson factorization is considered state-of-the-art in considering users' and items' implicit features. It considers users in terms of latent preferences, and items in terms of latent attributes. It also helps in recommending low activity and low rated items [28, 5, 29]. Matrix factorization based models gained attention after the Netflix challenge when Koren et.al [4] won the challenge. This model considers the ratings as a product of user and item hidden features. This model also considers the past temporal consumption effects of individual users. But this model didn't consider items' evolving attributes over time. Following this, some models proposed factorization of user-item-time tensor [14, 13]. Instead of modelling the evolution of users and items over time, their models quantify them by considering them as factor models over time. Some researchers proposed user-item time evolution by considering it as a Gaussian state-space model [30]. Some researchers considered Dirichlet distributed latent factors [31]. These all previously mentioned models are based on explicit information. But there are two more models following the same line, which focus on user item implicit data [32, 33]. These models consider users' implicit behavior, such as user click behavior. Researchers in [32] proposed a hidden Markov model which selects users' click-behavior. In another work, a gamma process dynamic Poisson factor analysis model is proposed to factorise a dynamic count matrix [33].

2.1. Data and knowledge sources

There are many different types of data and knowledge sources for recommendation systems. Whether the recommender is utilising that data depends on the approaches being employed. There are many different types of data that can be collected, including information about products, users, their social media activity, the psychology of the consumer's particular society, information about their geography, etc. However, in general, data is divided into three categories: products, users, and transactions.

2.2. Items

Items are the recommended objects by the system; Adomavicius *et al.* [34] notes that they can be identified by their utility or value. An item's value can be positive or negative, depending on whether the user finds it beneficial or whether the user chose poorly when making the purchase. The customer will pay a price when they purchase an item. The item has two costs: one is the cognitive cost of looking for and choosing an item, and the other is the actual price users pay for the thing.

2.3. Users

Different users of an RS may have various traits and objectives. To tailor the recommendations, RS needs to use a variety of user-specific data. In a collaborative filtering method, users are modelled as a simple list that contains the ratings provided by comparable users for particular objects. Age, gender, profession, and education are RS socio-demographics that are utilised in demographic analysis. The user model is made up of user data. There are numerous methods for user modelling that can be used, and in this regard, a recommendation system can be seen as a tool that creates recommendations by developing and utilising user models[35]. A user can also be described by behavioural pattern data, e.g. web browsing activity in a web or online recommender, or travel destination search query patterns in a travel recommender system[36].

2.4. Transactions

Transactions are the data that is generated during a user's interaction with the system. It is also known as log data. A recommender system algorithm exploits these transactions for generating recommendations. Transaction data can be the user's explicit rating or the user's selection of items from a list of items. Each record in the database captures a transaction such as a customer's purchase, a flight booking, or a user's click on any item [37]. But most commonly, ratings are being used; the rating may be explicit or implicit. In explicit rating, the user is asked to provide a rating, like in Amazon's five-star book rating. Ratings can be in a variety of forms.

- Numerical rating such as Amazon's five-star rating.
- Ordinal rating like: strongly agree, agree, neutral, disagree, strongly disagree.
- Binary rating like good or bad.
- Unary rating can be observed as the user purchases an item. In implicit ratings, the system aims to infer ratings from the actions taken by the user. For example, implicit rating is considered by considering a user's implicit behaviour such as clicking, the time duration of visiting an item, and so on.

2.5. Recommendation Techniques

In order to predict an item using the aforementioned fundamental attribute, the system must execute specific actions. There is a choice between personalised and non-personalized recommendations. A personalised RS provides recommendations that are specific to the user, as opposed to a non-personalized system, which makes recommendations based on both items and past behaviour. A new user will then be recommended based on both the behaviour of the item and the user's previous behaviour if a piece of music is liked and rated by the users. The collaborative filtering approach is one of its better uses. The recommendation uses essentially five main kinds of strategies:

1. Content-Based recommendation
2. Collaborative filtering
3. Hybrid Method
4. Multidimensional Recommendation Model
5. Graph-Based Recommendation

2.5.1. Content Based Recommendation

In content-based, the system should recommend items that are similar to the given item if the user has liked the item in the past. The learning algorithm considers items' features for calculating the similarity of the item [34].

The utility $\Omega(u, p)$ of item p for user u is estimated based on the utilities $\Omega(u, p_i)$ assigned by user u . that are similar to item p [34]. The content-based recommendation is based on information retrieval and information filtering research. But the current content-based recommendation system is more than just information retrieval. The advancement over traditional information retrieval comes from the fact that it is also considered a user profile that contains information about the user's interests, preferences, and habits as well. This information may be collected explicitly by asking the user or implicitly from transactional data.

Let P be the set of items features collected in the keywords form. Then this set will be used, using the user's profile and making predictions. e.g., in web page recommendation on every page, there will be keywords and its frequencies, let k_i is the word in d_j document then its weighting measure w_{ij} can be calculated in many ways. One of the best-known techniques is the term frequency/inverse document frequency (TF-IDF).

If N is the total number of document that can be used for recommendation, with keyword k_i appears in n_i of them and if f_{ij} is the frequency of keyword k_i in document d_j then TF_{ij} , the term frequency on normalize frequency of keyword k_i in document d_j is :

$$TF_{i,j} = f_{i,j} / (\max_z f_{z,j}) \quad (1)$$

Where the maximum is calculated over the frequencies $f_{z,j}$ of all keyword k_z that appears in the document d_j . However, the keyword that appears in many documents is not useful in distinguishing between relevant and irrelevant documents. In the case of movie recommendations, some researchers applied to content based on the similarity of the movie poster and found good results.

2.5.2. Collaborative filtering based recommendation

In collaborative filtering systems, the user will be recommended items based on what people with similar interests and preferences liked in the past. The similarity in the tastes of two users is calculated based on the similarity in their rating history. That is why it is also called the "people to people correlation". Mathematically the utility $\Omega(u, p)$ of item p from user u is estimated based on the utility $\Omega(u_j, p)$ assigned to item p by those user $u_j \in U$ who are similar to user u [34]. For example, in a book recommender to recommend a book, the system will search for the user who has bought similar kinds of the book in the past, then the given user is most likely to like the book as those who have similar interests. Algorithms for collaborative filtering can be classified in two ways, memory-based or heuristic-based, and model-based. Memory-based algorithms use heuristics to predict unknowns from the given data. In model-based, from the given data, the system creates a model and then gives a value for an unknown value, like a probabilistic approach. There are two probabilistic models that have been reported by the research community [34]: cluster model and the Bayesian networks based RS model [38].

2.5.3. Hybrid method based recommendation

In Hybrid RS, two or more techniques are combined to make a recommendation, as some techniques work better in some contexts while others do not, like cold start problem, which can be partially solved if collaborative filtering is combined with content-based recommendation. Therefore, hybrid recommenders also perform well in some situations where we combine many techniques.

2.5.4. Multidimensional recommendation model

G. Adomavicius et al., [39] given the concept of multidimensional model, instead of classifying recommender system into many types (i.e. content, collaborative, demographic, community, knowledge based and hybrid [36]), they classified recommender system into only 4 types i.e content based, collaborative filter based, hybrid and Multidimensional Model(MD). According to the MD model, a recommendation model consists of three main components: multiple dimensions; a profile for each dimension; and rating aggregation capabilities.

This multiple dimension is the same as data mining and OLAP applications. Formally let $D_1, D_2, \dots, D_n$ be dimensions, each dimension D_i being a subset of a Cartesian product of some attributes or fields A_{ij} , ($j = 1 \dots K_i$) i.e. D_i is subset of $A_{i1} \times A_{i2} \times \dots \times A_{iK_i}$ where each attribute defines a domain (or set of values). Besides one or many attributes form a key attribute i.e the rest of the attributes can be easily identified by key attribute/s [39].

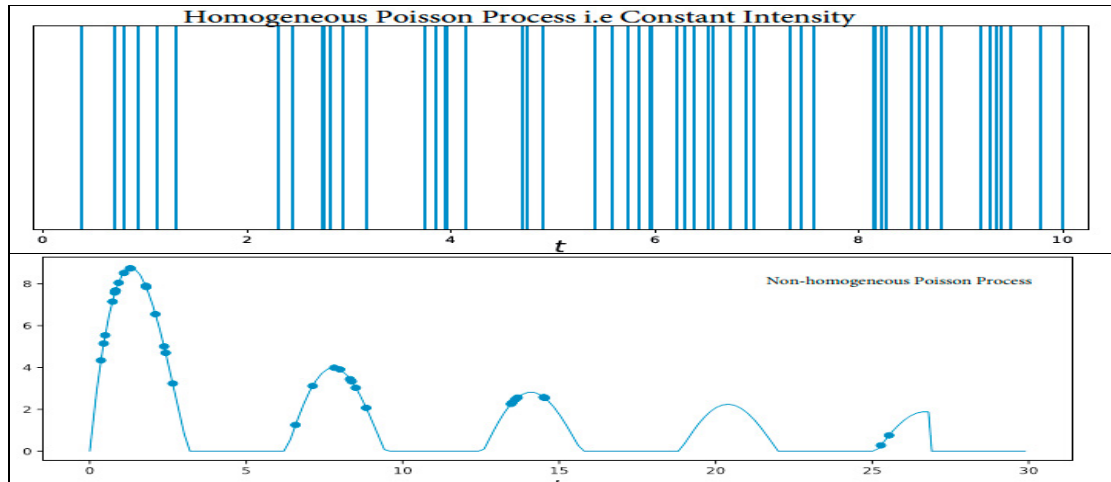


Fig. 2. This figures shows the intensity of homogeneous and non-homogeneous poisson processes.

2.5.5. Graph based recommendation

Real world phenomena can be effectively represented as a graph, which is also known as the graph-based knowledge representation problem. Researchers have been exploring it for the last couple of decades, and it has emerged as an important model for studying complex multi-relational data [40]. This knowledge base stores data in RDF triples (subject, predicate, and object). Researchers are exploring the possibilities of building a knowledge-graph inspired product graph [41]. The graph-based approaches are applied in the recommendation system and solved as a missing link prediction problem between user and item or user and user [42, 43, 44, 45, 46, 47, 48, 49]. In this framework, a bipartite graph data structure is used to represent the user-item interaction relation. The recommendation is made by the missing link prediction approach between the user and item [50], or ranking based [51, 52]. Graph-kernel based link prediction in bipartite graphs [53], Graph-based Clustering [47]. Some researchers have implemented a bipartite graph based recommendation in real systems with 3+ Billions of items and 200+ million [44]. They have found it more effective than the user-item matrix based implementation, which suffers from a sparsity problem. An example of graph-based recommendation using deep learning (Autoencode)-

3. METHODOLOGY

The problem of personalized, time sensitive recommendations can be solved by using point process-based models. A point process is a random collection of points falling into the same space, such as an occurrence of an event on the timeline. Let n sequence be represented by an ordered list of their occurrences on a time-line event, $evnt(t_i) = t_0, t_1, \dots, t_{n-1}$.

$$x(t) = \sum_{i=0}^{n-1} \Theta_{t,t_i}, \quad (2)$$

where Θ_{t,t_i} is a function which outputs 1 at time t_i event have happened other wise 0.

3.1. Poisson Process

The temporal Poisson process is a stochastic process which is effectively applied for natural dynamics prediction. According to the Poisson process, the probability that n events occur within some closed interval $[0, T]$ follows a Poisson distribution-

$$P(n) = \frac{e^{-\lambda T} (\lambda T)^n}{n!}, \quad (3)$$

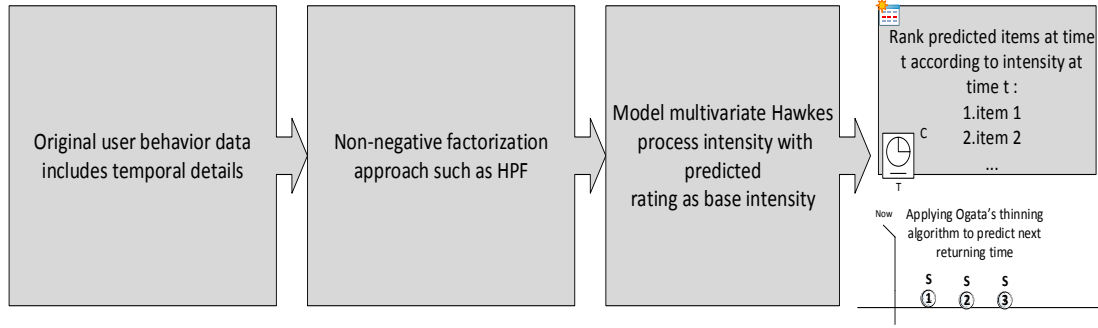


Fig. 3. A framework for time sensitive recommendation.

where λ is known as intensity rate with mean λT . These events are supposed to be completely independent, therefore heavily applied in predicting various evolutionary processes. In Poisson process modeling, it considers the time stamped events according to the interval in which they occur and therefore can estimate the future intensity of the occurrence more precisely. Poisson processes can be categorised into two different processes based on the processes generated by the underlying systems. 1) Homogeneous Poisson process, in which intensity of generating process remains constant and 2) Non-homogeneous Poisson process, in which intensity of generating process changes over time. Most of the real world systems fall into the second category, which is why non-homogeneous systems are of more interest to researchers. In Fig.2 we have shown the intensities of two processes. Another kind of process is also known as the Hawkes process, which is a combination of homogeneous and non-homogeneous Poisson processes. It is found that the superposition of different Poisson processes is also a Poisson process. The main difference between Poisson and Hawkes is that in the former case it is considered that events are independent, while in the latter case it is considered that the future evolution depends on the whole history, which makes it a self exciting process. In our study, we will use the Hawkes process.

3.2. Poisson Factorization(PF)

Poisson factorization [54] is a general framework for making a recommendation which exploits a user's implicit feedback, for example, snaps, buys, or watches. Based on PF, researchers have presented Hierarchical Poisson Factorization (HPF)[55] which utilises the occasions a user u interacted with the item p to construe the enthusiasm of a user for all items, including non-interacted ones. PF is the essential model in this system, which expects that on the occasions that user u or clicks an item, p is a Poisson arbitrary variable r_{up} with rate λ_{up} . HPF models the enthusiasm of user u and properties of item p as K dimensional latent vectors θ_u and β_p , individually, and utilize their internal product as the latent rate of r_{up} , that is:

$$r_{up} \sim \text{Poisson}(\theta_u^T \beta_p) \quad (4)$$

3.3. Proposed framework: Hierarchical Poisson Factorization with Cyclic Behaviour (HPFCB)

In this section we technically defined our proposed framework, which can be easily understood from Fig.3. Let we have user transaction history $\bar{h}(T)$ of user item interaction event up to time T . Then $\bar{h}(T)$ can be represented as a set of events e_i , formally $\bar{h}(T) = \{e_i\}_{i=1}^{\#(T)}$, where $e_i = (u_i, p_i, t_i)$ is the triplet of user u , item p and time $t \in T$; T is the set of all interaction events when user u interacted with item p . $\#(T)$ is the total number of interactions up to time T . $\bar{h}_{up}(t)$ denotes the set of interaction e_i until time t from user u on item p . N_u is the set of users which are followed by user u . This relation is from users' social network, apart from the user item interaction graph.

It is obvious that a user interacts with various items depending on two driving forces, the first of which are intrinsic user preferences and the second of which is prior experiences that the user's friends have had with the item. For instance, a user can listen to a music depending on his own interests or when he discovers that his friends have loved and shared it on social media. The duration of user involvement with various items is a collection of interdependent

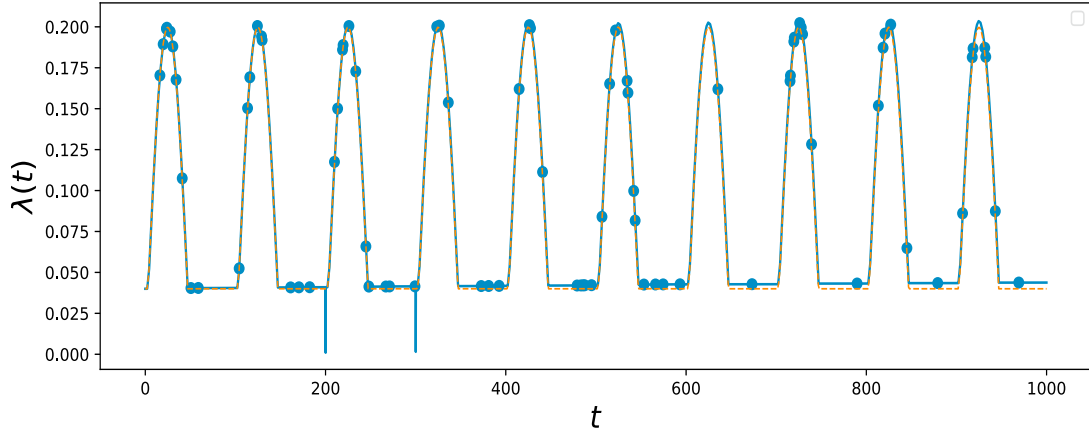


Fig. 4. Results on synthetic data to show how effectively Hawkes process are able to fit cyclic pattern.

Poisson processes in HPFCB models. One of the characteristics of the Poisson process is that it may be described as a superposition of separate Poisson processes[56]. In HPFCB, we believe that each process's intensity is composed of two elements. Extrinsic intensity is the inclination of a user to interact with an object based on previous interactions, which may also be cyclic, and intrinsic intensity is based on user preferences and an item's features. We employed a hierarchical Poisson process to calculate the intrinsic intensity [55]. We also took into account the intensity, which takes cyclical behaviour into account, while computing the extrinsic component. The following are the values of the combined HPFCB process intensity, λ :

$$\lambda_{up}(t) = \theta_u^T \beta_p + \sum_{e \in h(t)} \Psi_{u,p}(e, t) \quad (5)$$

where,

$$\Psi_{u,p}(e, t) = \{1 - r_0 \sin(\frac{2\pi}{T_m}(t + \delta))\} e^{-(\frac{t-t_e}{\gamma})} A_{up}(t) \quad (6)$$

where, r_0 is the relative oscillation of the cycle, δ is the phase t_e is the time when user u interacted with item p and γ is the decay of influence of item p on user u . $A_{up}(t)$ is the binary adjacent matrix which have 0 or 1 entries. 1 shows if user u have consumed item p at time t_e . And θ_u, β_p are Gamma distributed random variables [55]. Our model is basically Hawkes process with homogeneous intensity ($\theta_u^T \beta_p$) and non-homogeneous intensity ($\Psi_{u,p}(e, t)$). As it is found that the Hawkes process can effectively detect cycles, for example, see our results on synthetic data in Fig. 4. We have generated the results as described in the paper [57]. Our kernel is also inspired by the cyclic behaviour of Twitter users'[58].

3.3.1. Recommendation using HPFCB

a) To make a recommendation we infer the posterior and make recommendation according to the expected intensity of the user to engage with different items. To achieve this, we take the expectation of our intensity, sort the items in descending order, and recommend the sorted item to the user.

b) To predict the returning time of user u and item p we sample the returning time from Ogata's thinning algorithm [59](See Algo.3.3.1) and estimate the returning time by sample average, the same as [6].

Ogata's thinning algorithm [1] *Input:* $\lambda_{up}(t)$, last session start time of user $t_{s_0}^u$, upper bound on time T , $t = 0, S = \{\}$
 $\beta \leftarrow \max(\lambda_{up}(t)) \forall t \in [t_{s_0}^u, T]$ Sample next arrival time $s \leftarrow e^{1/\beta}$ Generate Random Number $r \sim \text{Uniform}([0, 1])$ Set $t \leftarrow t + s$ $j \leftarrow \text{patlen}$ $r \leq \lambda_{up}(\frac{t+s}{\beta})$ $S = S \cup t$. S .

$$E[\lambda_{up}(t)] = E[\theta_u]^T E[\beta_p] + \sum_{e \in h(t)} E[\Psi_{u,p}(e, t)] \quad (7)$$

4. Experimental Setup

We test our proposed model HPFCB's effectiveness using a real-world data-set. We demonstrate that HPFCB effectively captures changes in trends and user interests that other temporal models are unable to, in addition to modelling recurrent user behaviours throughout time. We have reported the results after 10 experiments.

4.1. Data acquisition

At present, the most feasible available data sets for recommendation systems are from online open repositories such as user activity, for example, movies, songs, videos, social media and so on. But to consider our problem, we have song listening behaviour data, *Last.FM* [60]. This dataset contains the records of 1200 users' listening behaviour. It contains 1000 artists, which eventually generates 48000 events. This dataset has sufficient user recurrent activity, which can be used as a benchmark.

4.2. Evaluation Metrics

We have used the following metric to evaluate our model.

Normalized discounted cumulative gain (NDCG): The NDCG works on all the items in the ranking list. NDCG algorithmically decays as a function of its rank. It can be formulated as-

$$\frac{1}{N} \sum_{u=1}^N \sum_{\alpha \in Y_u^L} \frac{1}{\text{rank}(u, \alpha)} \quad (8)$$

where, Y_u^L is the set containing user u 's test observation and N is the number of users. $\text{Rank}(u, \alpha)$ is the predicted rank for item α for user u [5].

4.3. Results and comparison

In figure [4] we have shown how hawks process able to fit the cyclic pattern. Further we have used the real world data set to evaluate the performance of our proposed model. In figure [5] we have shown the result size of top k recommendation with our benchmark (HPF). We reported the Normalized Discount Cumulative Gain (NDCG) between the true parameters of the model and the estimated ones. NDCG works on ranking items in true and compares it with the predicted ranking set of items. As it can be seen initially when number of items in the recommendation list is low then the performance of out proposed model as well as the benchmark HPF model is low. As the number of items size increases the performance of our model as well as benchmark both increases. In other words they show similar trends with respect to recommendation list size k . Over all our proposed model out performs the benchmark with high margin.

5. CONCLUSIONS

In this study, a unique framework that we offer, can be used to create recommendation systems from implicit feedback data. In order to accomplish this, a Hawkes process based model is proposed, which accounts for the cyclic behaviour of user activity such as listening to music during a specific time of day by using a personalised initial intensity based on Hierarchal Poisson Factorization and a dynamic activity by taking into account sinusoidal function coupled with exponential decay of influence. Empirical results demonstrate that the proposed model outperforms a state-of-the-art model in terms of prediction accuracy. In addition, our model is more interpret-able than other deep neural network based models, which can be easily tuned according to business need. The use of generative deep neural networks to model the intensity functions is another intriguing area for future research.

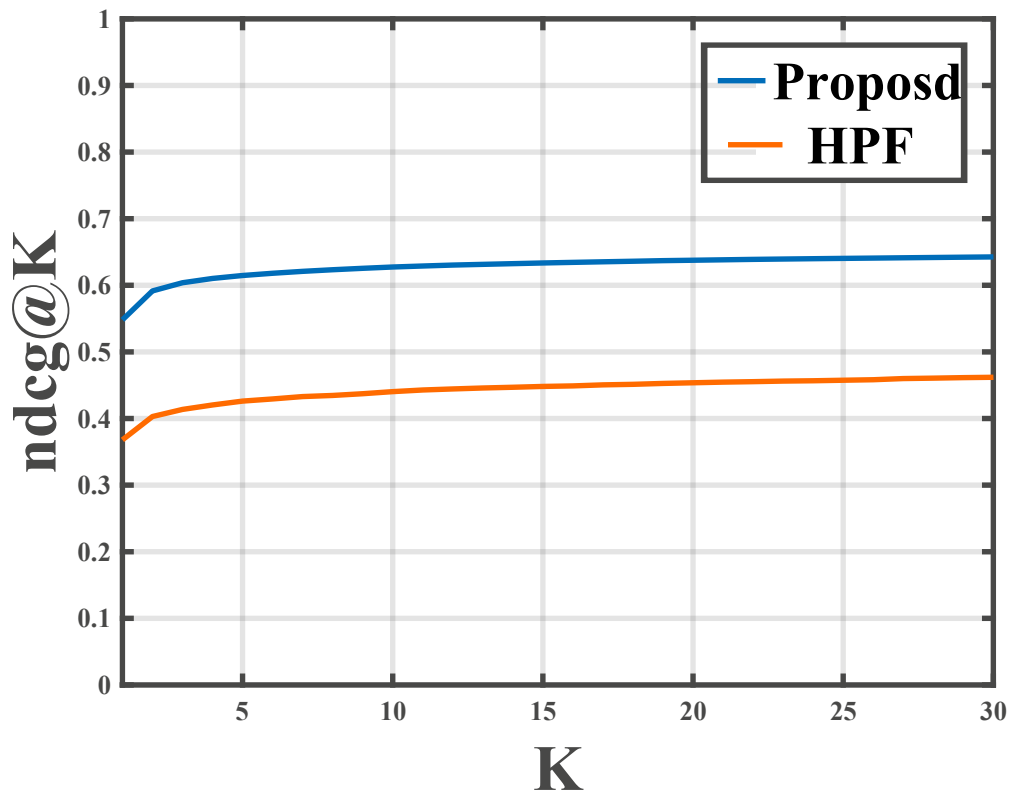


Fig. 5. NDCG@k results of our proposed method and benchmark model on Last.FM data set. The X-axis shows the top k recommendation list size

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